

ICE LOW SULPHUR GASOIL FUTURES (G)

Spread & Curve Structure -- Fundamental Reference

The European Diesel Benchmark, ARA Logistics, and Refining Margins

This document presents a fact-based reference on the fundamental drivers of ICE Low Sulphur Gasoil futures (ticker: G), contract size 100 metric tonnes, the benchmark price reference for diesel and gasoil traded in Europe and a key global distillate benchmark alongside NYMEX Heating Oil (covered in a separate reference document in this series). The contract prices low sulphur gasoil (10ppm diesel) delivered into the ARA region (Amsterdam-Rotterdam-Antwerp, including Flushing and Ghent) -- Europe's principal refined products trading and storage hub, playing a structurally similar role to the US Gulf Coast for NYMEX-priced products. Gasoil differs from its NYMEX Heating Oil counterpart in one critical demand-side respect: while the US distillate barrel is split between heating oil (a winter heating-fuel demand pillar with no significant European equivalent at this contract's scale) and on-road diesel, the European gasoil barrel is overwhelmingly a transport-fuel product, reflecting Europe's historically much higher passenger and commercial vehicle diesel penetration relative to the United States. This document presents the gasoil-specific fundamentals in full, with particular emphasis on the ARA delivery and storage hub, the European refining and import balance, the structural demand differences from US heating oil, and the HO/LSGO spread connecting the two distillate benchmarks.

Futures First -- Internal Education Material
For Analyst Group Use

1. Contract Specifications and Delivery Mechanics

ICE Low Sulphur Gasoil is a physically delivered contract for ultra-low sulphur (10ppm) diesel/gasoil, delivered via barge, small coaster, or in-tank/inter-tank transfer within the ARA hub -- a delivery mechanism reflecting the contract's role as the pricing reference for a deep, liquid regional barge market rather than a single-port delivery system.

Parameter	Detail
Exchange	ICE Futures Europe (London)
Ticker	G (front month); the contract is formally named "Low Sulphur Gasoil Futures," commonly referred to in the trade simply as "Gasoil" or "LSGO"
Underlying	Low sulphur gasoil, 10ppm diesel specification -- ultra-low sulphur diesel meeting EU fuel quality standards
Contract size	100 metric tonnes per lot; delivery by volume is 118.35 cubic metres per lot (the volumetric equivalent of 100 metric tonnes of 10ppm diesel)
Quotation	US dollars per metric tonne (USD/MT)
Contract months	Monthly contracts listed for an extended forward curve (materially more granular than the quarterly-cycle grain and livestock contracts covered elsewhere in this series, consistent with the monthly-cycle convention used throughout the energy complex)
Settlement type	Physical delivery -- seller to buyer, into barge (or coaster up to 15,000 dwt) or via in-tank/inter-tank transfer from an Exchange-recognised bonded storage installation or refinery in the ARA area, on a day nominated by the buyer within a mutually agreed 5-day delivery range between the 16th and the last calendar day of the delivery month.
Delivery location	Amsterdam-Rotterdam-Antwerp (ARA), including Flushing and Ghent -- Europe's principal refined products storage and trading hub
Quality verification	Quantity and quality verified by one Exchange-approved inspector, selected from two nominated by the buyer -- a documentary/inspection-based delivery verification process distinct from the warehouse-receipt systems described for the soft commodity contracts elsewhere in this series.
Last trading day	12:00 hours, 2 business days prior to the 14th calendar day of the delivery month
Contract value at \$700/MT	\$70,000 per contract
Related instruments	Heating Oil/Low Sulphur Gasoil (HO/LSGO) futures spread, directly tradeable on ICE; American-style options on the futures, delta-hedged against the underlying G contract.

1.1 The ARA Hub -- Europe's Refined Products Centre of Gravity

- **What ARA is:** the Amsterdam-Rotterdam-Antwerp region (extended to include the nearby Flushing and Ghent terminals) is Europe's largest refined products storage, blending, and trading hub, anchored by Rotterdam's position as Europe's largest port and a major independent storage tank farm cluster, alongside significant refining capacity at and near the hub.
- **ARA's structural role:** the hub functions for European refined products broadly as the US Gulf Coast functions for NYMEX-priced products -- the deep, liquid physical market against which the futures contract settles, with extensive independent storage capacity allowing the hub to absorb supply and demand imbalances and providing the inventory data most closely watched by gasoil market participants (Section 6).

- **Barge market liquidity:** physical gasoil trading in ARA is conducted predominantly via barge transactions moving product within the Rhine-Scheldt river and canal network connecting the hub to inland European demand centres (particularly Germany, via the Rhine), making barge freight rates and river conditions (addressed in Section 7.2) a relevant logistics variable, broadly analogous in function to the Mississippi River barge system's role in US grain and product logistics described elsewhere in this series, though serving the inland distribution of refined products rather than agricultural commodities.

2. Gasoil vs Heating Oil -- The Structural Demand Difference

Although ICE Gasoil and NYMEX Heating Oil price chemically similar middle-distillate products and are directly linked via the tradeable HO/LSGO spread (Section 8), the demand composition underlying each contract differs in one structurally important respect that should inform any fundamental analysis of gasoil specifically.

2.1 Europe's Diesel-Dominant Transport Fuel Mix

- **Historically high diesel penetration in European passenger vehicles:** European passenger car fleets have, for several decades, had a substantially higher diesel-engine share than the US passenger fleet, a divergence rooted in historical European fuel taxation policy that favoured diesel relative to gasoline, and in European vehicle manufacturers' corresponding investment in diesel engine technology. This gives gasoil demand a transport-fuel weighting with no precise equivalent in the NYMEX Heating Oil contract's demand base, as described in the Heating Oil reference document in this series.
- **The post-"Dieselgate" diesel share decline:** the European passenger diesel share has, since the 2015 emissions-testing scandal commonly referred to as "Dieselgate" (and the subsequent regulatory and reputational consequences for diesel passenger vehicles), declined from its historical peak, with new vehicle registrations shifting toward gasoline and, increasingly, electric and hybrid powertrains. This represents a documented structural headwind for the passenger-vehicle component of European gasoil demand specifically, distinct from the commercial/heavy-goods diesel demand pillar described below, which has not experienced a comparable shift.
- **Commercial and heavy goods vehicle (HGV) diesel demand:** unlike the passenger segment, diesel remains the dominant fuel for European HGV and commercial freight transport, given diesel engines' favourable torque and fuel-efficiency characteristics for heavy-load applications -- a demand pillar with no significant near-term substitution pressure comparable to the passenger segment, and one directly tied to European industrial production and freight activity levels (Section 5).

2.2 No Significant European Equivalent to US Heating Oil Demand

- As described in the Heating Oil reference document, a meaningful share of the US Northeast specifically still relies on heating oil for residential and commercial space heating -- a demand pillar with strong winter seasonality. European residential heating relies overwhelmingly on natural gas, electricity, and district heating networks rather than oil-fired heating systems at anything approaching the scale of the US Northeast, meaning gasoil demand lacks the equivalent winter heating-driven seasonal demand pillar that shapes a portion of the NYMEX Heating Oil seasonal pattern -- a structural difference that should inform any attempt to directly transpose Heating Oil seasonal patterns onto the Gasoil contract.

3. European Refining and the Diesel Supply Gap

A structural feature of the European refining system, longstanding and well documented, underpins much of the gasoil market's fundamental picture: European refineries, configured historically around gasoline-and-diesel-balanced crude slates and processing technology, have not produced enough diesel/gasoil to meet European demand for an extended period, making Europe a structural net importer of diesel.

3.1 The European Refining Configuration

- **The diesel/gasoline imbalance:** because European vehicle demand has historically skewed toward diesel (Section 2.1) while European refineries' crude processing and conversion unit configuration produces a gasoline/diesel output ratio that does not naturally match this demand skew, Europe has, for many years, run a structural diesel deficit alongside a gasoline surplus -- the mirror image of the US refining system's gasoline-weighted demand and diesel-export profile described in the RBOB reference document in this series, where the US exports a substantial share of its distillate output.
- **European refinery rationalisation:** European refining capacity has, over the past two decades, undergone significant consolidation and closure, driven by structurally weak refining margins on the gasoline side (given limited European gasoline demand growth and competition from larger, more complex refineries elsewhere) and by the broader European energy transition policy environment -- a trend that has, at the margin, further widened the structural diesel supply gap by reducing aggregate European refining capacity.

3.2 Import Dependence -- Sourcing the Diesel Deficit

- **Historical Russian diesel imports:** prior to 2022, Russia was a major supplier of diesel/gasoil to the European market, reflecting both geographic proximity and the historical configuration of Russian refining capacity toward diesel-weighted output suited to European demand.
- **The EU import ban on Russian refined products (February 2023):** as part of the broader sanctions response to Russia's invasion of Ukraine, the EU implemented a ban on seaborne imports of Russian refined petroleum products (following an earlier crude oil import ban), eliminating what had been a major source of European diesel supply and forcing a documented, significant reconfiguration of European diesel import sourcing toward alternative suppliers.
- **Alternative sourcing -- the Middle East, Asia, and the US:** following the Russian import ban, European diesel import sourcing shifted toward the Middle East (particularly new, large-scale refining capacity additions in Saudi Arabia and the UAE), India (which has expanded export-oriented refining capacity substantially), and, to a lesser degree, the United States -- each entailing materially longer shipping routes and higher freight costs than the prior, geographically proximate Russian supply, a structural change in the European diesel import logistics picture that traders should treat as an ongoing feature of the post-2022 market rather than a temporary disruption.
- **Suez Canal and Red Sea routing as a relevant logistics variable:** given the reliance on Middle Eastern and Asian diesel imports following the Russian ban, disruptions to Suez Canal and Red Sea shipping routes (including the documented Houthi attacks on commercial shipping in the Red Sea beginning late 2023) are a directly relevant logistics and freight-cost variable for European diesel import flows specifically, a consideration that did not carry the same weight for European gasoil before the post-2022 sourcing shift.

4. Global Production and Trade Flows

Understanding the gasoil/diesel market requires tracking not only European refining and demand but the broader set of exporting regions that supply the European import deficit described in Section 3.

Region	Role in Gasoil/Diesel Trade	Key Structural Facts	Gasoil (G) Relevance
Middle East (Saudi Arabia, UAE, Kuwait)	Rapidly growing diesel exporter, increasingly significant European supplier	Substantial new, large-scale, complex refining capacity additions over the past decade (including major Saudi and UAE projects), configured with significant diesel/distillate yield	A primary post-2022 replacement source for former Russian volumes; Middle Eastern refinery outage or maintenance schedules are a relevant European supply variable
India	Major and growing export-oriented diesel refiner	Large, complex, export-oriented refining capacity (notably including very large private-sector refineries) configured substantially for export rather than purely domestic Indian demand	A significant alternative diesel source for Europe following the Russian import ban; Indian refinery turnaround schedules and domestic Indian demand growth both affect export availability
United States	A diesel exporter on net, given the gasoline-weighted demand structure described in the RBOB reference document	US Gulf Coast refining capacity, with distillate yield calibrated significantly toward export markets including Europe and Latin America	A secondary but relevant source of transatlantic diesel flows to Europe, competing for the same export barrel with Latin American and other destinations
Russia	Formerly a major European supplier; now largely excluded from the EU market by the 2023 import ban (Section 3.2)	Continues exporting diesel to non-EU destinations (including, per various trade-flow analyses, redirected flows toward Asia, Turkey, and other non-sanctioning markets), but the direct EU import channel is closed	No longer a direct European gasoil supply source, though Russian export redirection affects the broader global diesel balance and, indirectly, the price of diesel barrels competing for the same end markets that Europe also draws on
China	A variable net exporter, subject to domestic export quota policy	Chinese refined product export volumes are managed via a government export quota system, making Chinese export availability a policy-administered rather than purely market-driven variable	A secondary global supply variable; Chinese quota policy changes affect the broader Asian and global diesel balance that indirectly influences the price of cargoes competing for European import demand

5. Demand Fundamentals

5.1 European Road Transport Demand

- As described in Section 2.1, HGV and commercial freight diesel demand is the more structurally resilient component of European gasoil demand, directly tied to European industrial production indices and freight activity volumes -- making Eurozone Manufacturing PMI and freight-sector activity data (analogous in relevance to the US ISM and freight data described elsewhere in this series) a relevant demand-side input.
- **Passenger diesel demand trajectory:** as described in Section 2.1, the passenger vehicle diesel share faces an ongoing structural decline driven by post-Dieselgate consumer/regulatory sentiment and the broader European shift toward electric vehicle adoption, itself significantly influenced by EU vehicle emissions regulation (including CO2 fleet emissions targets) and member-state EV purchase incentive policy -- a multi-year structural demand headwind distinct from any cyclical demand variation.

5.2 Agricultural and Off-Road Diesel Demand

- Gasoil (in a lower-tax, dyed/marked form in many European jurisdictions, sometimes referred to as "red diesel" or agricultural diesel, distinct from the road-tax-paid product priced by the ICE futures contract but drawn from the same underlying refined product pool) is also used extensively in European agricultural machinery, construction equipment, and other off-road/non-transport applications, giving the broader gasoil demand pool some exposure to European agricultural and construction sector activity cycles in addition to the on-road transport demand described above.

5.3 Industrial and Power Generation Backup Demand

- Gasoil serves as a backup fuel for industrial processes and power generation in some applications, with demand from this channel rising during periods of natural gas price spikes or supply disruption (as documented during the 2022 European natural gas crisis, when elevated gas prices increased the relative attractiveness of gasoil-fired backup generation and some industrial fuel-switching where technically feasible) -- a secondary but documented cross-fuel substitution demand channel linking gasoil to the European natural gas price environment.

6. ARA Inventory Data and the Physical Market

As with US product inventory data (EIA weekly stocks, described in the Heating Oil and RBOB reference documents), ARA independent storage inventory data is the primary physical-market tightness signal for European gasoil.

6.1 Independent Storage Data Providers

- **Insights Global / PJK International weekly ARA stock data:** published weekly, covering independently-held storage tank inventories of gasoil/diesel (and other refined products and crude) at the ARA hub -- the closest European analogue to the EIA's weekly petroleum status report, though sourced from independent commercial data providers rather than a government statistical agency, a structural difference from the US data infrastructure worth noting explicitly.
- **Interpreting ARA stock data:** rising ARA gasoil stocks generally signal a well-supplied or oversupplied near-term market (bearish for front-month gasoil and for backwardation in the curve), while falling stocks signal tightening (supportive for front-month pricing and backwardation) -- the same general interpretive framework described for US product stocks, applied to the ARA-specific data series.

6.2 Refinery Maintenance ("Turnaround") Season

- European refinery planned maintenance ("turnaround") seasons -- concentrated, as in the US, in the spring and (to a lesser degree) autumn shoulder seasons between peak gasoline-driving-season and peak winter-heating-demand periods -- reduce European refining output and can tighten the gasoil market if maintenance is concentrated or if unplanned outages coincide with the scheduled turnaround season, a seasonal supply-side pattern broadly analogous to the US refinery maintenance season described in the Heating Oil and RBOB reference documents.

7. The Forward Curve Structure

Gasoil's forward curve reflects the standard cost-of-carry framework for storable refined products, modulated by the European refinery turnaround calendar, the post-2022 import sourcing pattern described in Section 3.2, and freight/shipping conditions.

7.1 Carry and Backwardation

- Normal-market contango reflects storage, financing, and insurance costs for holding physical gasoil at ARA. Backwardation signals near-term physical tightness -- typically reflecting low ARA stocks, strong HGV/freight demand, or an import-side disruption (a Middle Eastern refinery outage, a Red Sea shipping disruption affecting the timing of incoming cargoes, or -- historically -- a Russian supply disruption prior to the 2023 import ban taking full effect).

7.2 Freight and Shipping Route Considerations

- **Rhine barge freight and water levels:** as described in Section 1.1, inland European gasoil distribution relies substantially on Rhine barge transport; low Rhine water levels (a recurring European summer/autumn weather risk, with notably severe and well-documented episodes in 2018 and 2022) can constrain barge loading capacity and disrupt the movement of product from ARA to inland demand centres, particularly Germany -- a documented logistics risk with some structural similarity to the Mississippi River low-water risk described for US agricultural logistics elsewhere in this series, here applied to refined product distribution.
- **Suez/Red Sea routing:** as described in Section 3.2, the post-2022 shift toward Middle Eastern and Indian import sourcing has made Suez Canal and Red Sea shipping conditions a directly relevant freight-cost and delivery-timing variable for European gasoil imports specifically, a consideration of increased structural importance relative to the pre-2022 market.

8. The HO/LSGO Spread -- Connecting the Two Distillate Benchmarks

Because ICE Gasoil and NYMEX Heating Oil price chemically similar middle-distillate products in two different currencies, units, and regional markets, the spread between them is a directly tradeable, exchange-listed relationship and one of the most important cross-market signals for either contract.

8.1 The HO/LSGO Futures Spread Contract

- **Unit conversion mechanics:** because Heating Oil is quoted in US cents per gallon and Gasoil in USD per metric tonne, the ICE HO/LSGO Futures Spread contract converts gasoil into a price per gallon using a published conversion factor of 312.9 (derived from the metric-tonne-to-barrel conversion of 7.45 and the barrel-to-gallon conversion of 42), allowing the two products to be compared and spread-traded on a like-for-like basis.
- **Contract ratio:** the minimum HO/LSGO spread quantity is 4 lots, comprising 3 Heating Oil lots and 4 Low Sulphur Gasoil lots -- a ratio reflecting the relative contract sizes of the two underlying futures (NYMEX Heating Oil at 42,000 US gallons per contract versus ICE Gasoil at 100 metric tonnes per contract) needed to approximate equivalent notional exposure on each side of the spread.

8.2 What the HO/LSGO Spread Represents

- The HO/LSGO spread captures the relative tightness of the US versus European distillate markets -- a widening spread in favour of one region signals relatively tighter conditions there (and, mechanically, creates an arbitrage incentive for distillate cargoes to be redirected toward the relatively higher-priced region, within the bounds of shipping economics and time), a relationship broadly analogous in concept to the cross-exchange arbitrage relationships described for copper (COMEX-LME) and cocoa (CC-XC) elsewhere in this document series, here operating through actual physical cargo redirection rather than financial arbitrage alone.
- **Demand-composition differences as a driver of the spread:** because the two markets' demand composition differs as described in Section 2 (US winter heating oil demand having no European equivalent; European HGV diesel demand and the post-Dieselgate passenger diesel decline having no precise US equivalent), seasonal and structural demand shifts specific to one region but not the other are a recurring source of HO/LSGO spread movement distinct from pure global crude-price-driven moves.

9. Macro and Cross-Market Drivers

9.1 Crude Oil -- Brent as the Primary Input Cost

- As a refined product, gasoil's price is fundamentally anchored to crude oil input costs, with ICE Brent (rather than NYMEX WTI) the more directly relevant crude benchmark given gasoil's European/Atlantic Basin pricing context -- the gasoil crack spread (gasoil price less the Brent-equivalent crude cost) is the key refining margin metric, conceptually equivalent to the gasoline and heating oil crack spreads described in the RBOB and Heating Oil reference documents, here calculated against Brent rather than WTI given the contract's European market context.

9.2 European Natural Gas Prices

- As described in Section 5.3, elevated European natural gas prices can increase gasoil demand at the margin through fuel-switching and backup generation demand, making the TTF (Title Transfer Facility) European natural gas benchmark price a relevant secondary cross-commodity input, particularly during episodes of acute European gas market stress.

9.3 US Dollar

- As a globally traded, USD-denominated commodity, gasoil exhibits the standard dollar-priced-commodity relationship described elsewhere in this document series, with the added consideration that European buyers transact in EUR domestically while the futures contract and underlying physical cargoes are USD-denominated, making EUR/USD a relevant basis consideration for European end-users and refiners hedging via the G contract.

9.4 Speculative Positioning

- **ICE Commitments of Traders-equivalent positioning data:** as an ICE Futures Europe-listed contract, Gasoil is subject to the UK/EU market position-reporting framework rather than the US CFTC's Commitments of Traders regime that applies to NYMEX Heating Oil -- the same regulatory-regime distinction described for the CC/XC cocoa, No.11/XW sugar, and SO3/SR3 rates pairs elsewhere in this document series applies here in an analogous way.

10. Documented Seasonal Patterns and Historical Events

10.1 Seasonal Patterns

- **Spring/autumn refinery turnaround seasons:** as described in Section 6.2, reduced European refining output during concentrated maintenance periods can tighten the gasoil market, particularly if turnarounds are concentrated or coincide with unplanned outages.
- **Absence of a sharp US-style winter heating-demand seasonal pattern:** as established in Section 2.2, gasoil lacks the pronounced winter seasonal demand spike that characterises a portion of the NYMEX Heating Oil market, meaning gasoil's own seasonal pattern is comparatively flatter and more driven by the refinery maintenance calendar and broader crude oil seasonal patterns (described in the Brent and WTI reference documents) than by a distillate-specific demand season of its own.

10.2 Documented Historical Events

- **2022 European energy crisis and gasoil fuel-switching:** the sharp rise in European natural gas prices following Russia's invasion of Ukraine and the subsequent reduction in Russian pipeline gas flows to Europe increased gasoil demand at the margin via the fuel-switching and backup generation channel described in Section 5.3 and Section 9.2, contributing to elevated gasoil prices and crack spreads during this period.
- **February 2023 EU ban on Russian refined product imports:** as detailed in Section 3.2, this sanctions measure eliminated a historically major source of European diesel supply, forcing a documented, structural reconfiguration of European import sourcing toward the Middle East, India, and the United States, with longer shipping routes and correspondingly different freight-cost and timing dynamics than the prior Russian supply pattern.
- **Late 2023-ongoing Red Sea shipping disruptions:** Houthi attacks on commercial shipping in the Red Sea, beginning in late 2023, affected vessel routing for cargoes transiting the Suez Canal -- a documented logistics disruption of increased relevance to European gasoil imports specifically given the post-2023 reliance on Middle Eastern and Asian supply sources described in Section 3.2, with affected vessels in some cases rerouting via the Cape of Good Hope, adding significant transit time.
- **2018 and 2022 Rhine low-water episodes:** as referenced in Section 7.2, documented periods of low water on the Rhine constrained barge transport of gasoil and other products from ARA to inland German and other European demand centres, widening regional basis and disrupting normal distribution logistics.

11. Key Data Sources and Release Schedule

Report / Data	Publisher	Frequency	Typical Timing	Primary Gasoil Curve Impact
ARA independent storage stock data	Insights Global / PJK International	Weekly	Variable, typically early week	Front contract -- the primary European physical tightness signal, analogous to EIA stocks for US products
ICE Brent futures and crack spread	ICE Futures Europe / market data providers	Continuous	Market hours	All contracts -- primary input-cost benchmark and refining margin signal
Platts/Argus European diesel cargo and barge assessments	S&P; Global Platts; Argus Media	Daily	Market close	Front contracts -- physical market price assessments underlying much of the basis and cash market
Eurozone Manufacturing PMI	S&P; Global / HCOB	Monthly	~1st-3rd business day	C2-C6 -- HGV/freight diesel demand proxy via industrial activity
EU new vehicle registration data (diesel share)	ACEA -- European Automobile Manufacturers Association	Monthly	Variable	Structural demand trend -- tracks the ongoing passenger diesel share decline described in Section 2.1
TTF European natural gas price	ICE Endex / market data providers	Continuous	Market hours	Secondary demand input via fuel-switching, particularly relevant during gas-price-spike episodes
EU sanctions/import policy announcements (Russian refined products)	European Commission	As scheduled/ amended	Variable	All contracts -- direct effect on European import sourcing availability
Suez Canal/Red Sea shipping advisories	Shipping industry sources; Suez Canal Authority	As events occur	Variable	Front-to-mid contracts -- freight cost and import timing for Middle Eastern/Asian-sourced cargoes
Rhine water level data	German Federal Waterways and Shipping Administration	Daily during risk season	Continuous	Basis-level -- inland distribution logistics, particularly relevant in summer/autumn low-water periods
EU refinery capacity/closure announcements	Company disclosures; trade press	As announced	Variable	Back curve -- structural European refining capacity trajectory
ICE Futures Europe position-reporting data	ICE Futures Europe / relevant EU regulator	Periodic per regime	Variable	Positioning / crowding -- under the UK/EU framework distinct from US CFTC COT

ICE Low Sulphur Gasoil prices the European diesel market from a structurally distinct demand and supply position relative to its NYMEX Heating Oil counterpart: a transport-fuel-dominated demand base (HGV freight as the resilient pillar, passenger diesel in structural post-Dieselgate decline) with no significant winter heating-demand equivalent, layered onto a structural European diesel supply deficit that has, since February 2023, required a fundamental reconfiguration of import sourcing away from Russia and toward the Middle East, India, and the United States -- a shift that has materially increased the relevance of Suez Canal and Red Sea shipping conditions to the European gasoil balance.

For spread traders, the most reliable framework tracks ARA independent storage data as the primary physical tightness signal, the European refinery turnaround calendar as the dominant seasonal supply pattern (in the absence of a sharp demand-side season of gasoil's own), the post-2022 import sourcing geography and its associated freight/shipping risk as a structural rather than temporary market feature, and the HO/LSGO spread as the primary cross-market relationship connecting the European and US distillate markets -- a relationship whose movements are frequently driven by the structural demand-composition differences between the two regions described throughout this document, not solely by global crude price moves.